

Term 4 Capstone: Full-Stack IoT Dashboard

Complete Lesson Guide & Student Handbook — Connecting, Monitoring, & Automating Real-World Systems

1. Introduction: Solving Real Problems with IoT

In modern communities, daily life presents critical challenges that require real-time monitoring and swift action:

- **Water Overflow:** Overhead water tanks in homes often overflow, wasting thousands of liters of clean water and electricity.
- **Farm Irrigation:** Agricultural crops suffer when fields receive too much or too little water, leading to crop failure or water wastage.
- **Air Quality Alerts:** Major urban centers (like New Delhi) frequently experience critical Air Quality Index (AQI) spikes, threatening public health.

What is IoT (Internet of Things)? IoT is the network of physical devices embedded with sensors, software, and connectivity technologies that connect and exchange data with other devices and systems over the Internet. By building *smart dashboards*, we can see, understand, and automatically resolve these challenges in real time.

Aarav's Tip: "Sensors collect data, dashboards show it — that's IoT magic!"

2. IoT Architecture: The 3 Core Layers

An end-to-end (Full-Stack) IoT system works seamlessly by dividing tasks into three interconnected layers:

Layer Name	Primary Function	Real-World Examples
1. Sensors / Perception Layer	Senses and collects physical data directly from the environment.	Ultrasonic water level sensors, Soil Moisture probes, MQ air quality sensors.
2. Network & Cloud Gateway Layer	Transmits collected data wirelessly from local devices to cloud servers.	Wi-Fi modules, cellular gateways, cloud messaging protocols.
3. Application & Dashboard Layer	Displays data visually and provides user controls and automation rules.	Interactive dashboards with gauges, trend line graphs, status indicators, and toggles.

Ananya's Tip: "Each layer has a special job! It's just like sending a message through the internet!"

3. Step-by-Step Guide: Building Your Smart Dashboard

Step 1: Layout & User Interface Design

Design an intuitive interface using a drag-and-drop UI builder. Select appropriate visual widgets based on the data type:

- **Gauges:** Best for bounded numerical ranges (e.g., Water Tank Level 0–100%).

- **Line Graphs:** Ideal for tracking continuous changes over time (e.g., Air Quality Index or Temperature).
- **Switches/Toggles:** Used for manual control actions (e.g., Turning a Water Pump ON or OFF).

Step 2: Data Binding (Connecting Sensors to Widgets)

Link each physical or simulated sensor feed to its corresponding dashboard widget. When the sensor reports a new measurement, the UI updates dynamically without refreshing the page.

Step 3: Status Indicators & Visual Alerts

Incorporate color-coded status indicators (e.g., green for normal, red for critical) to enable rapid visual assessment:

- **Green Light:** Normal conditions (e.g., Safe water levels, healthy soil moisture, safe AQI).
- **Red Light / Alert:** Critical thresholds reached (e.g., Tank near overflow, hazardous air pollution level).

4. Automation Logic: IF-THEN-ELSE Rules

To make an IoT system truly "smart," we implement rule-based logic using block programming. Automation removes the need for manual human intervention.

```
IF Water Level is HIGH
THEN Set Pump Status to OFF
ELSE Set Pump Status to ON
```

5. Testing, Debugging, and System Presentation

Testing & Debugging Workflow:

1. **End-to-End Simulation:** Trigger changes in sensor values and confirm that dashboard widgets update instantaneously.
2. **Automation Verification:** Verify that relays/switches respond automatically when threshold conditions are triggered.
3. **Debugging Steps:** If data fails to reflect, systematically check physical wire connections, Wi-Fi connectivity, MQTT/HTTP cloud feeds, and visual widget data-bindings.

Student Practice Assessment

Term 4 Capstone: Full-Stack IoT Dashboard — Knowledge Check

Practice Questions

Answer the following questions based on the full-stack IoT lesson. (Note: Answers are not provided. Complete these on a separate sheet or in your workbook.)

Question 1 (Multiple Choice)

Which layer of the IoT architecture is directly responsible for measuring physical phenomena such as soil moisture or water level?

- A) Application Layer
- B) Network & Cloud Gateway Layer
- C) Sensors / Perception Layer
- D) User Interface Layer

Question 2 (Short Answer)

Explain the function of the Network/Cloud Gateway layer in an IoT system. How does data travel from physical sensors to an online user dashboard?

Question 3 (Dashboard Design & Application)

Select the most appropriate dashboard widget (e.g., Gauge, Line Graph, Switch) for each of the following scenarios, and justify your choice:

- a) Displaying the temperature trends over a 24-hour period.
- b) Manually turning a farm irrigation pump on or off.
- c) Indicating the current fill percentage of an overhead water tank.

Question 4 (Logic & Automation)

Write an **IF-THEN-ELSE** block logic statement to automate a farm irrigation system based on a Soil Moisture Sensor. Specify when the irrigation motor should turn ON and when it should turn OFF.

Question 5 (Troubleshooting & Debugging)

Imagine your dashboard gauge is not updating when the physical water sensor is submerged in water. List three systematic debugging steps you would take to find and fix the problem.

Question 6 (Real-World Impact Essay Question)

Choose one real-world problem (Water Overflow, Farm Irrigation, or City Air Quality). Describe how implementing a full-stack IoT dashboard system can benefit your local community or environment.